

R version 4.4.0 (2024-04-24) -- "Puppy Cup"
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Platform: aarch64-apple-darwin20

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Natural language support but running in an English locale

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Type 'demo()' for some demos, 'help()' for on-line help, or
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Type 'q()' to quit R.

[R.app GUI 1.80 (8376) aarch64-apple-darwin20]

[History restored from /Users/alperkaragol/.Rapp.history]

```
> rm(list = ls())
> x0 <-
c(8.1354,7.7566,6.3782,4.4443,4.199,4.4659,4.3355,5.1443,5.1539,5.3589,4.5494,3.966,4.5133,4.4266,2.7235,2.6853,3.6399,
2.3924,2.1805,3.1697,3.251,3.5719,2.4666,3.1838,3.8558,3.3107,2.8778,3.8824,2.4669,2.1032,2.1848,2.1262,2.1197,1.6936,0
.972,1.1911,1.2167,1.3261,1.4047,1.4025,1.2111,1.0978,0.975,1.2058,1.1489,1.3001,1.3729,1.3415,1.2484,2.8054,2.6505,1.9
035,2.6381,2.6071,2.6928,2.1778,2.8493,3.1658,2.9118,3.3506,4.9982,4.4456,2.6249,2.0404,2.8236,3.917,1.0022,4.4704,4.72
15,3.8634,5.4015,0.9197,1.7351,1.538,1.4984,1.1971,1.213,1.0317,1.094,1.5545,1.1457,0.5209,0.6673,0.6336,0.4708,0.7104,
1.0608,0.1305,0.5242,0.8218,1.4169,1.5236,0.9245,0.7492,0.2881,0.4435,0.4381,0.7294,0.9776,0.6536,1.0279,1.1539,0.6147,
0.5569,0.6307,0.6464,0.893,0.6432,0.3539,0.2637,0.5732,0.9026,0.8754,0.759,1.0612,1.6178,1.5088,0.9649,1.3695,1.8251,1.
7146,1.6661,1.052,1.1221,0.8858,0.5957,0.5596,0.9722,0.8616,0.6078,0.6724,0.2838,0.4854,0.2775,0.3775,0.4977,0.2211,0.6
656,1.2225,1.432,0.7009,0.9805,0.9532,0.659,0.566,0.4233,0.2085,0.2489,0.6105,0.575,0.9155,1.076,0.794,0.5983,1.2308,0.
8728,1.8796,1.5093,0.3138,0.8097,1.2649,1.0342,0.9878,1.2006,0.4819,0.9141,1.3545,1.0996,0.8461,1.024,1.2344,1.3546,1.1
105,1.2248,1.9985,1.6244,1.8888,2.7608,3.1528,3.0861,3.3795,3.8718,4.541,4.7533,4.6,5.1642,5.1477,4.7961,5.0519,4.4039,
3.2859,2.9199,3.0052,3.4713,2.4658,3.1157,2.615,2.8623,2.4879,2.229,1.5157,1.7118,2.0976,1.7816,1.3475,2.0789,2.6503,1.
4095,1.0085,1.2308,1.3826,1.1455,1.3163,1.2408,0.7651,0.8604,1.1838,0.8769,1.1572,1.0945,1.1974,0.4625,1.1499,1.5805,1.
5959,3.2424,3.2052,2.6546,2.9587,4.1345,4.2244,4.2261,3.0107,1.0671,1.113,1.9708,1.3407,0.7922,1.3799,0.5167,1.3681,2.2
442,1.6753,1.8926,2.8769,3.2837,2.8158,3.2778,3.9995,3.6529,3.225,4.6815,5.3507,5.3315,5.4393,5.3367,4.8394,5.1685,3.73
39,3.7014,3.4954,3.6866,1.8403,3.1703,2.5396,4.1841,4.1809,4.2733,4.4203,5.0622,9.6973,10.833,12.709,6.9191,7.2805,9.41
49,3.0605,4.0946,5.2827,5.9893,7.4624,7.1254,5.8847,5.6955,5.8638,5.2168,4.2174,4.3437,3.6007,3.2393,3.0483,2.7008,2.22
75,1.9748,2.2016,2.3201,1.13,1.3085,1.1632,0.3162)
> x1 <-
c(1.5344,1.661,0.7362,1.3389,2.0589,2.8972,3.0469,3.1873,2.4545,1.2315,1.8267,1.571,1.5847,1.0503,1.6592,1.3515,1.0216,
1.1328,1.6074,1.0891,0.8391,1.5525,1.5141,1.4774,2.1025,2.6,2.6495,2.8334,3.1268,3.742,3.8461,4.3075,4.3298,4.9038,5.02
24,5.3824,5.4212,4.6784,5.0195,5.4907,5.3458,5.0929,5.3713,5.3433,4.8384,4.9319,5.0504,5.4166,6.0741,5.8553,5.477,7.321
,9.6859,11.978,15.286,12.718,14.598,14.652,15.813,18.649,20.493,18.854,18.824,16.636,16.131,13.662,13.07,11.651,9.9106,
7.009,8.4412,8.2876,10.216,10.775,13.257,14.494,16.508,17.349,19.422,20.199,18.951,19.033,17.119,18.195,17.031,15.017,1
3.781,11.738,9.6335,6.9841,8.0517,7.8721,9.8214,10.002,11.925,12.894,15.064,15.947,18.339,19.709,18.554,18.546,17.018,1
7.815,17.008,15.249,13.526,11.256,9.2855,6.7444,7.7229,7.1581,8.9692,8.5732,10.781,12.211,14.594,15.436,17.926,19.075,1
7.549,18.012,16.646,17.778,17.005,15.234,13.211,12.419,11.588,10.995,7.923,8.1664,9.4838,9.4923,9.8926,11.616,13.811,14
.758,17.126,18.513,17.579,17.961,16.154,17.768,17.33,15.333,14.874,15.56,16.333,12.932,7.2125,6.3819,4.9318,9.0908,15.1
62,16.582,13.052,7.4969,8.4289,7.2557,5.3203,5.0908,4.9724,4.537,3.444,3.1748,3.3543,3.6318,2.8356,2.8925,3.3827,2.5065
,2.2278,2.8302,3.2616,4.2184,4.4208,3.2119,2.9611,4.3246,3.9803,3.3534,3.727,4.4542,4.1844,3.7547,4.5126,4.5208,3.7899,
3.8098,2.9613,2.9532,2.8552,2.8183,2.4776,2.2201,2.3916,2.2283,1.7673,2.4097,2.9343,2.7168,2.5928,2.7329,3.4901,3.2622,
3.2639,4.3455,4.4021,5.8427,6.2822,8.3618,9.4197,10.74,9.7349,9.4411,8.5452,6.8972,6.8824,6.6269,5.4255,4.9061,5.0509,4
.7142,3.9112,3.6826,3.8802,2.8043,1.6748,1.9485,1.9321,1.9532,1.8244,1.2728,1.2873,1.3693,1.1537,0.9345,1.1838,0.4628,0
.3144,0.9387,1.1113,1.1349,1.5825,1.5425,1.9042,2.0371,0.9939,1.0986,1.1789,1.0826,0.74,1.1257,0.8684,0.7931,0.4627,1.1
292,1.6577,1.4752,1.484,2.0731,2.2299,2.2081,2.5851,3.0399,2.8652,3.6323,4.3159,4.2644,4.4377,3.7812,5.0069,6.5922,6.26
82,9.8051,12.961,10.124,8.5579,7.2512,7.08,5.3106,4.98,5.3042,4.7778,3.2264,2.8008,3.0487,2.9123,1.7475,1.9372,2.4121,2
.0823,2.0565,2.7831,2.4965,2.4904,3.6825,2.8134,2.3984)
> x2 <- c(3.444,3.225,2.6559,3.8291,4.362,2.8624,1.3547,0.8398,1.7248)
> x <- c(x0,x1,x2)
> y0 <-
c(1,2,3,2,4,3,4,5,3,4,5,4,6,6,8,8,6,9,9,9,4,9,5,7,8,9,8,9,8,6,8,9,9,4,9,7,4,8,7,9,6,9,9,9,9,9,8,9,9,7,8,9,9,6,8,8,8,9,9
,9,9,5,9,8,9,5,8,5,7,5,8,8,8,9,9,7,5,9,7,5,7,9,9,4,9,9,5,3,7,9,7,5,8,9,7,4,9,9,7,6,4,9,4,4,5,8,5,4,8,6,4,8,6,6,5,9,9,9,
8,9,5,9,5,4,9,8,4,7,9,5,2,8,9,4,9,9,9,6,7,9,8,9,8,7,6,7,7,9,8,9,5,8,8,9,9,9,9,9,9,9,9,8,9,9,8,9,5,9,8,8,9,9,8,7,8,9
,9,7,9,9,9,9,9,8,9,8,8,6,7,8,6,9,9,9,9,9,9,8,6,7,8,7,6,9,5,6,3,7,7,5,8,8,4,6,8,9,9,9,9,9,7,4,9,8,4,7,8,9,9,9,8,
8,9,7,9,8,4,7,9,9,9,9,9,8,9,8,9,4,8,9,5,9,9,7,9,4,3,8,5,5,4,5,1,6,5,4,5,5,4,4,4,5,3,4,1,3,3,5,1,1,3,4,1,1,3,3,1,1,4
,4,1,1)
> y1 <-
c(5,3,1,2,5,3,3,3,3,3,4,3,4,5,2,4,6,4,4,6,5,4,5,5,7,4,8,7,8,9,9,8,9,9,9,9,9,9,7,5,5,9,3,5,4,8,4,6,4,4,6,1,5,1,2,6,2,5
,3,5,4,5,6,9,8,8,7,7,9,9,4,8,2,4,5,1,4,3,5,6,5,8,3,3,4,5,4,7,5,8,4,8,5,7,7,4,9,4,4,8,3,9,6,6,6,5,8,4,4,7,1,9,4,8,9,8,5
,3,5,1,8,4,7,8,8,5,3,1,3,6,5,4,3,5,1,4,3,9,5,5,7,1,4,4,8,4,8,9,5,5,6,4,5,5,6,3,7,8,9,7,7,9,8,9,8,9,9,9,9,9,9,9,9,9,9
,9,9,7,9,9,9,7,9,9,7,6,9,9,7,6,7,8,7,7,9,9,5,7,9,7,9,4,8,8,8,7,9,7,7,9,6,8,8,8,7,8,9,6,5,9,9,8,9,7,8,5,8,8,9,9,9,9,7,8,7,
9,6,9,7,9,7,7,9,7,5,8,9,5,9,9,8,9,7,8,9,7,4,7,8,4,9,8,8,5,8,8,6,9,9,6,8,7,9,5,7,8,8,6,4,9,9,7,6,9,8,3,8,9,7,1,9,9,6,6,7
,7,8,9)
```

```
> y2 <- c(9,7,9,8,4,6,7,9,8)
> y <- c(y0,y1,y2)
> cor.test(x, y, alternative = "two.sided", method = "spearman", exact=FALSE )
```

Spearman's rank correlation rho

```
data: x and y
S = 47122653, p-value = 2.927e-10
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
-0.2517865
```

```
> # ---- Confidence interval ----
> if(!"RVAideMemoire" %in% installed.packages()){install.packages("RVAideMemoire")}
> library(RVAideMemoire)
*** Package RVAideMemoire v 0.9-83-7 ***
> spearman.ci(x,y)
```

Spearman's rank correlation

```
data: x and y
1000 replicates

95 percent confidence interval:
-0.3296628 -0.1757418
sample estimates:
      rho
-0.2517865
```

```
>
```